Traffic Congestion Prediction Using Categorized Vehicular Speed Data



Manoj Kumar and Kranti Kumar

Abstract Traffic congestion has become a problem which is associated with our day-to-day life. Industrialization, urbanization, and the increasing population of cities have the main role in this problem. Transportation agencies in almost all the countries are trying their best to alleviate the traffic congestion problem. This study develops a traffic congestion evaluation method, which is categorized into four states named low, medium, congestion, and serious congestion. Convolutional long short-term memory (Conv-LSTM) neural network (NN) was trained to learn multivariate features as the inputs like categorized vehicular speed, time, and day of the week with respect to the traffic speed as an output of the whole stream. Trained model was used for short-term traffic speed prediction. Also, the prediction accuracy and stability of the Conv-LSTM NN has been compared with other neural network models, e.g., multi-layer perceptron (MLP), cascade forward back-propagation (CFBP), recurrent neural network (RNN), long short-term memory (LSTM) NN, and convolutional neural network(CNN). Results confirm that the Conv-LSTM NN achieves higher prediction performance than the compared models. Python 3.8.3 was used for programming purpose. Model code automatically selects 80% and 20% time intervals of the datasets in a random way for training and testing, respectively, from the traffic data. Furthermore, Conv-LSTM NN was combined with the developed congestion evaluation method and then the traffic congestion state was predicted. Study results confirm that the Conv-LSTM NN can be successfully applied for the prediction of traffic speed and traffic congestion status with non-homogeneous traffic in the Indian context.

Keywords Traffic congestion · Congestion index · Conv-LSTM neural network · Speed prediction

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1 Introduction

Traffic congestion affects the economy of a city and its environment as a direct and indirect effect [20]. A study conducted by the Council of Scientific and Industrial Research-Central Road Research Institute (CSIR-CRRI), Delhi, clearly suggests that commuters have to pay more for extra fuel consumption while driving on congested roads in Delhi. CSIR-CRRI preliminary estimate puts the wastage on fuel expenses at Rs 960 crore a day across India [7]. In another study done by CSIR-CRRI, it was found that poor upkeep encroachments cause traffic jams on internal roads in Delhi [6].

Traffic congestion directly affects an individual in the form of time loss, mental stress, increased travel cost, etc., whereas it affects the global level in the form of added pollution which leads to global warming. Traffic congestion prediction in advance provides the traffic management agencies, with the required time to divert the traffic on less congested routes or to take other suitable steps to make a smooth journey for travelers. Initial traffic forecasting models mainly focused on traffic parameter prediction such as flow, speed, and density under homogeneous traffic conditions. Most of the developed countries have homogeneous traffic and lots of research is available for homogeneous traffic prediction to mitigate traffic congestion [1, 21]. However, developing countries like India follow non-homogeneous (mixed) traffic. Research findings from homogeneous traffic can not be applied for mixed traffic scenarios, because vehicles do not follow proper lane discipline in a heterogeneous traffic stream. Traffic data collection is an important part to check the accuracy and reliability of developed models. Developed countries use modern equipment like road sensors, inductive loops, and installed video cameras to collect the real-world traffic data. However, these data collection equipment and techniques are costly [12].

Through the study of several models/algorithms on traffic prediction, it was noticed that most existing deep learning models [16, 25, 28] ignore the internal relationship between spatial and temporal information. Thus, the existing research seems insufficient for the prediction of traffic information in terms of traffic flow, speed, and congestion under mixed traffic conditions. This study presents a methodology to predict traffic congestion using the time series data of multi-locations. It predicts traffic congestion in four states namely low, medium, congestion, and serious congestion. Proposed traffic congestion states evaluation method is inspired by the method of Kong et al. [11]. In this paper, we applied the Conv-LSTM NN for traffic congestion prediction. The vehicular categorized data was employed to learn mixed traffic features and traffic speed was used as the target parameter. After training the model, target parameter from selected time steps was predicted. Then the congestion evaluation method was applied with the target parameter for the traffic congestion states forecast.

2 Literature Review

In this section, a brief overview is presented associated with ideas in the prediction of traffic flow, speed, and congestion. With the availability of big data, the development of artificial intelligence has led researchers to apply various models in this research area.

The autoregressive integrated moving average (ARIMA) model was applied to motorway data from France to predict traffic flow using previous traffic data [27]. Su et al. [24] proposed an incremental support vector regression (SVR) method for traffic flow prediction. A traffic flow prediction model named the Kalman filtering technique (KFT) was presented[13] and evaluated which can be applied to small input datasets. Hidden Markov model has been applied [30] for traffic congestion prediction during peak hours and its accuracy was checked with a neuro-fuzzy approach. MLP neural network was applied [14] for traffic flow prediction using past data collected from mixed vehicular traffic. Ma et al. [19] tested the effectiveness of LSTM NN with several traditional networks for speed prediction using the same real-time speed data collected from microwave sensors in Beijing. Bhatia et al. [3] implemented an LSTM architecture for real-time traffic flow prediction with an accuracy of 97% on the overall dataset. Gated recurrent units (GRU) NN and LSTM NN were used for traffic flow prediction and the experimental results demonstrate their better performance than ARIMA model [8]. Liu et al. [17] developed a deep learning model formed by combining CNN and LSTM NN, called Conv-LSTM for traffic flow prediction. A deep learning model was presented [9] to predict the traffic speed using RTMS sensors in Beijing. Boukerche and Wang [4] developed a deep learning model by combining the CNN and GRU models for traffic flow prediction. Mixed deep learning (MDL) model was introduced [18] for forecasting lane-level short-term traffic speed. MDL was constructed by using the Conv-LSTM layers. In the study of Inner Ring Road, Delhi, Rao and Rao [23] established the congestion thresholds on urban arterials to identify the traffic congestion in terms of speed using floating car data. Kong et al. [11] proposed a fuzzy evaluation method for traffic congestion estimation using trajectory data. Chakraborty et al. [5] used CNN, to detect traffic congestion from image-based data. Ranjan et al. [22] constructed a hybrid neural network by adding CNN, LSTM, and transpose CNN to predict the city-wide traffic congestion. A model named SG-CNN was proposed by Tu et al. [26] for traffic congestion prediction, which is inspired by the road segment grouping algorithm to optimize the training process. The aforementioned mentioned statistical methods like ARIMA models, nonparametric regression models, KFTs, and traffic flow theory-based models are primary models [1] which were used for traffic prediction problems. The statistical methods consider only time-series information of parameters, therefore these methods become inadequate to capture the changes in traffic parameters. To mitigate the drawbacks of statistical methods, various machine learning-based methods including SVR, k-nearest neighbor, locally weighted learning, and traditional neural networks (like MLP and CFBP, etc.) were applied. These models also appear weak to solve time-series problems having large data with variations. RNN was able

to learn the time-series information with the input of the previous time intervals from time-series data. However, RNN has failed to deal with gradient vanishing problems like remembering long-term historical information [2]. LSTM NN is known to deal with temporal problems over long sequence data. But, LSTM NN has some disadvantages: (1) it is challenging to use and (2) it is slow in processing when input and output sequences have a number of features. Similarly, CNN is known to deal with image processing problems by extracting the effective pixels from the image and it has been successful in object detection, segmentation, and image understanding. Unlike LSTM NN, CNN is capable of processing high-resolution multi-feature data because it has three abilities like local connectivity, weight sharing, and pooling. Thus, a combined network of LSTM and CNN i.e. Conv-LSTM can be successful to learn long-term time series data with a number of input features.

Till date, limited research exists associated with the deep learning approach to deal with traffic congestion prediction. The aim of this study is the accurate prediction of traffic congestion for short-term duration. In this article, Conv-LSTM neural network has been applied for traffic speed prediction. The network uses time-series data, which was collected at two different locations in Delhi city using video cameras. Then, the developed congestion evaluation method was applied to the predicted speed for the prediction of traffic congestion class.

3 Methodology

3.1 Study Data

Traffic data were collected from Majnu ka Tila, Delhi. Figure 1a shows the identified location. Selected location was having straight roads, clear site distance, and no traffic restrictions like intersections and mid-blocks. Outer Ring Road is one of the most important highways in Delhi. Its length is 47 km with a three-lane road. It connects Delhi areas as encircles like ISBT Kashmere Gate, Majnu ka Tila, Azadpur, Rohini, Vikash Puri, IIT Delhi Gate, Kalkaji, Okhla, and ISBT Kashmere Gate (Azadpur $\rightleftharpoons$ ISBT Kashmere Gate). Traffic data were collected from the foot over bridge at Majnu ka Tila using Sony Handycam 16.6 megapixels video cameras as shown in Fig. 1b. These cameras were fixed at a suitable height using camera stands. Data collection was done without disturbing the moving vehicles in the full traffic stream from 16 to 22 November 2020 (during 8:00 am–11:00 am in the morning peak and 4:00 pm-7:00 pm in the evening peak). Vehicles were classified into seven categories like Car/Van/Jeep, Motorcycle/Scooty, three-wheeler, Bus, Light Commercial Vehicle/Minibus/ambulance, Truck, and Cycle/Rickshaw. Data extraction was done using Kinovea window software from the recorded video film. Categorized vehicular travel time was calculated from a marked 30m road trap length and its corresponding speed was obtained by using speed, time, and distance relationship. For further analysis,

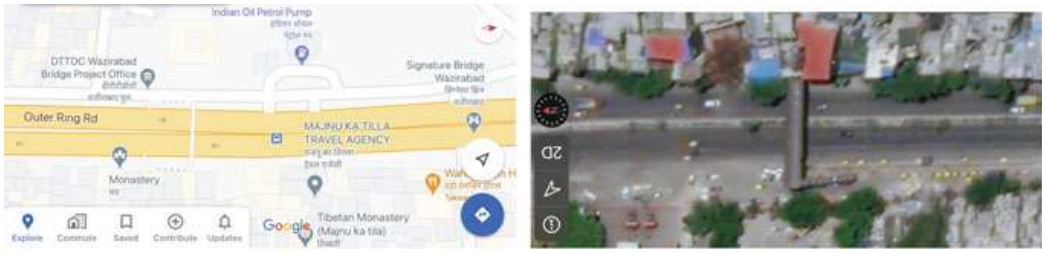


Fig. 1 Data collection locations with direction at the foot over bridge, Majnu Ka Tila, Delhi

Table 1 Summary of observed speed (in km/h) for different vehicle categories

	Intervals of 5 min				Intervals of 10 min			
Vehicle category	Min.	Max.	Mean	SD	Min.	Max.	Mean	SD
C/J/V	2.13	83.08	30.76	11.05	5.89	81.82	30.76	10.12
S/M	3.56	81.82	32.67	12.06	10.06	70.26	32.68	11.10
3W	3.08	69.23	28.07	10.38	4.79	51.92	28.08	9.33
Bus	3.77	63.16	22.23	8.16	4.89	56.25	22.24	7.37
LCV	2.35	67.50	25.38	10.03	4.07	70.34	25.61	9.66
Truck	1.66	61.02	22.13	7.59	4.52	71.76	22.29	7.72
Cycle	0.18	14.03	13.09	4.76	3.68	16.42	13.09	4.30
Average speed	4.46	49.56	24.96	7.24	7.97	43.46	24.97	6.91

[a] C/J/V—Car/Jeep/Van, S/M—Scooty/Motorcycle, 3W—Three-wheeler, LCV—Light commercial vehicle, Min—Minimum, Max—Maximum, SD—Standard deviation

extracted data in the intervals of 5 and 10 min were entered in an excel sheet for both directions.

As per location information, two datasets named input dataset and target dataset were prepared. Input datasets have 1008-time intervals and 504-time intervals (as rows) for time intervals of 5 min and 10 min, respectively, and 9 features (as columns) including time of day, day of the week, and categorized vehicular speed. Target dataset has 1008-time intervals and 504-time intervals (as rows) of 5 min and 10 min, respectively, and one features the average speed of categories vehicular speed. Variation of categorized vehicular speed is represented in Table 1. For all the models, previous data was used for predicting traffic speed in the upcoming 5 and 10 min of interval.

3.2 Conv-LSTM Neural Network

In this section, first, we give a brief introduction to LSTM NN and CNN. Then, we describe the structure of the Conv-LSTM model. To deal with the research gaps

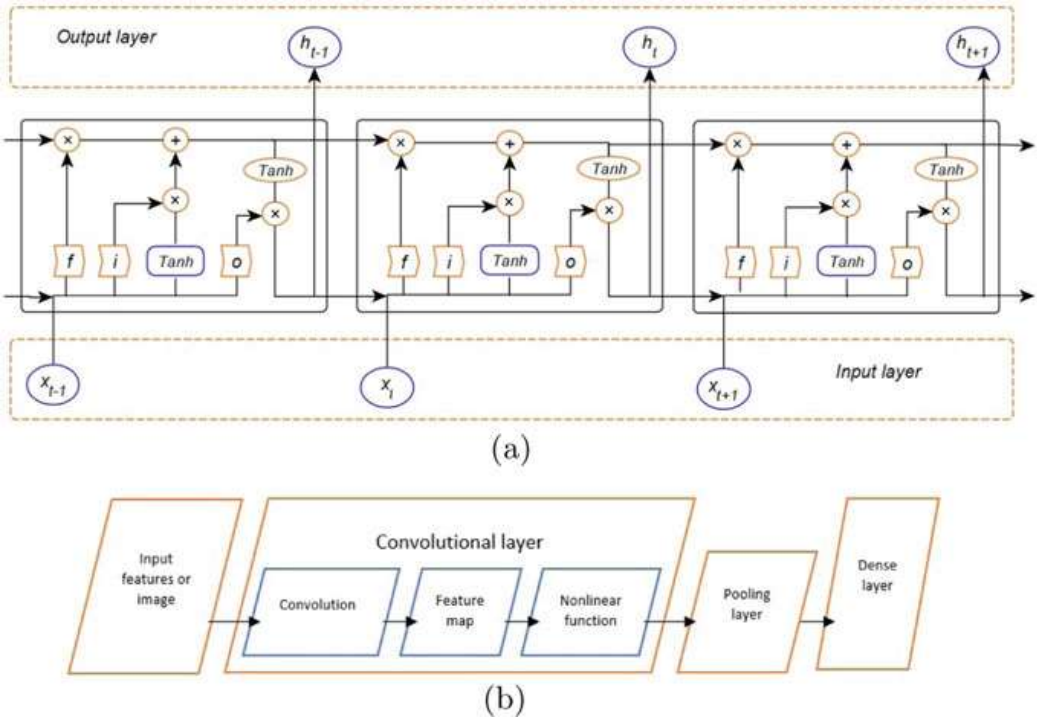


Fig. 2 Representation of structures **a** LSTM **b** CNN

described in the Sect. 2, LSTM neural network was originally proposed by Hochreiter and Schmidhuber [10]. LSTM neural network was proved more efficient than traditional RNN [19]. The hidden layer of the LSTM neural network is a memory block (LSTM block), which consists of the cell state c_t. The memory block also has three special gates, which are known as the input gate i_t, forget gate f_t, and output gate o_t. Due to the presence of gates and cell state, the output of LSTM NN gets influenced at every moment. These gates are activated using the Sigmoid activation function which produces the information in terms of a value lying in the interval of [0, 1], which can be understood from Fig. 2a. The network processes input sequence $x = (x_1, x_2, \ldots, x_N)$ by iterations from $n = 1$ to N in following equations (1)–(4):

$$i_t = \sigma(U_i x_t + V_i h_{t-1} + W_i C_{t-1} + b_i) \tag{1}$$

$$f_t = \sigma(U_f x_t + V_f h_{t-1} + W_f C_{t-1} + b_f) \tag{2}$$

$$g_t = \tanh(U_g x_t + V_g h_{t-1} + b_g) \tag{3}$$

$$o_t = \sigma(U_o x_t + V_o h_{t-1} + W_o C_t + b_o) \tag{4}$$

where Us, Vs, and Ws respectively are the weight matrices of input, recurring connections, and cell state, bs are the bias vectors. σ and $tanh$ are the Logistic (Sigmoid) and Hyperbolic tangent activation functions, which have been used for gates activation and cell activation, respectively. Along with these gates, the current internal cell state C_t and current hidden state (or output) h_t is defined as in the Eqs. (5) and (6).

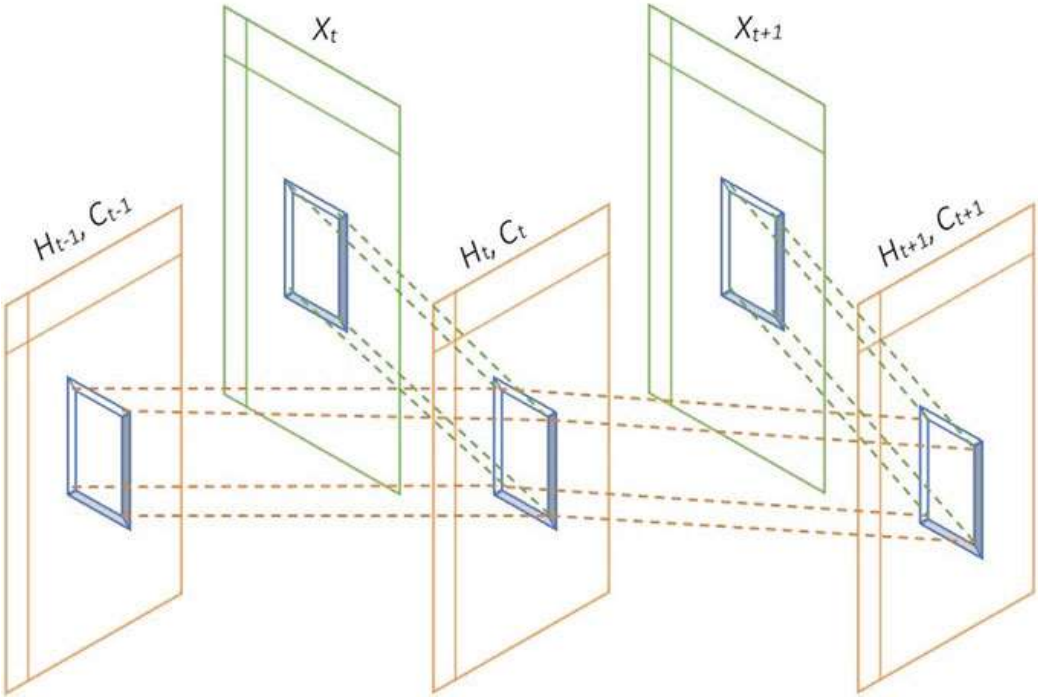


Fig. 3 Representation of Conv-LSTM NN

$$C_t = f_t \odot C_{t-1} + i_t \odot g_t \tag{5}$$

$$h_t = o_t \odot tanh(C_t) \tag{6}$$

During the training process, the weights and biases were initialized first. Afterward, the parameters were trained using back-propagation and gradient boost method was used to optimize the loss that occurred during this process. More description about the LSTM neural network can be found in [19].

The Conv-LSTM NN has the advantage of both CNN and LSTM. It also captures the spatial and temporal information of the traffic simultaneously. CNN is a type of deep learning network, which is specialized for feature extraction (or processing data). Detailed description about CNN can be found in [15]. The structure of CNN is shown in Fig. 2b. Conv-LSTM has similar processing equations to LSTM processing equations (1)–(6), but the variable dimension is different from the LSTM. $\mathcal{X}_1, \mathcal{X}_2, \ldots, \mathcal{X}_t$ are the input tensors, and their corresponding $\mathcal{C}_1, \mathcal{C}_2, \ldots, \mathcal{C}_t$ are cell states, and $\mathcal{H}_1, \mathcal{H}_2, \ldots, \mathcal{H}_t$ are hidden states. Note that $\mathcal{I}_t$, $\mathcal{F}_t$, and $\mathcal{O}_t$ are respectively three gates named input, forget, and output while all lie in $R^{P \times Q \times R}$ where P and Q respectively are the total rows and the total columns in the dataset; R is the length of each feature vector. The methodology of Conv-LSTM NN is expressed by Eqs. (7)–(11)

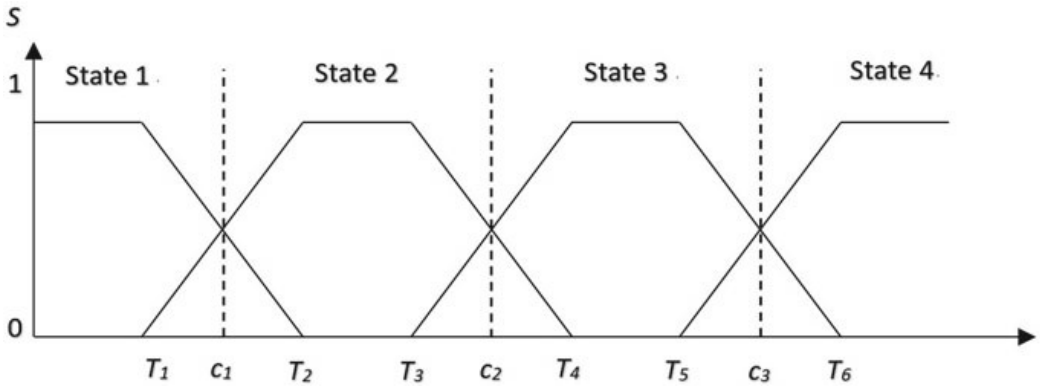


Fig. 4 Representation of membership function with congestion states

$$\mathcal{I}_t = \sigma_a(U_i \oplus \mathcal{X}_t + V_i \oplus \mathcal{H}_{t-1} + W_i \odot \mathcal{C}_{t-1} + b_i) \tag{7}$$

$$\mathcal{F}_t = \sigma_a(U_f \oplus \mathcal{X}_t + V_f \oplus \mathcal{H}_{t-1} + W_f \odot \mathcal{C}_{t-1} + b_f) \tag{8}$$

$$\mathcal{C}_t = \mathcal{F}_t \odot \mathcal{C}_{t-1} + \mathcal{I}_t \odot tanh(U_c \oplus \mathcal{X}_t + V_c \oplus \mathcal{H}_{t-1} + b_c) \tag{9}$$

$$\mathcal{O}_t = \sigma_a(U_o \oplus \mathcal{X}_t + V_o \oplus \mathcal{H}_{t-1} + W_o \odot \mathcal{C}_t + b_o) \tag{10}$$

$$\mathcal{H}_t = \mathcal{O}_t \odot tanh(\mathcal{C}_t) \tag{11}$$

where $\oplus$ represents the convolutional operator, $\odot$ represents the matrix multiplication; Us, Vs, and Ws are the weight tensors; bs are the bias tensors. Structure of Conv-LSTM is represented in Fig. 3 and more details about the network can be found in [18].

3.3 *Traffic Congestion Measure*

In this section, we derive a traffic congestion evaluation method, which is inspired by the fuzzy method. Developed congestion evaluation method was applied to generate evaluation sets in the form (low, medium, congestion, and serious congestion). In this method, first, membership function is determined along with the weight coefficients of the evaluation factor. Congestion evaluation method considers one or more traffic parameters as the evaluation factor. For simplicity and due to the availability of data, only one traffic parameter was considered, i.e., traffic speed. It is believed that the selected traffic speed parameter can characterize the traffic state accurately. Traffic congestion state was measured by exploring the traffic speed parameter. Identified traffic states play an effective role in the trips and traffic management system. Low, medium, congestion, and serious congestion categories were considered as the evaluation set.

To achieve this objective, the trapezoidal membership function was selected. It is a combination of piecewise linear and trapezoidal [29] as shown in Fig. 4, which can express knowledge of ambiguity caused by linguistic assessments by objectively

converting them into numerical variables. It determines the relationship between the evaluation factor (traffic speed) and the evaluation set for traffic congestion. Graph of membership function along with congestion states is shown in Fig. 4. In Fig. 4, $T_1, T_2, \ldots T_6$ are the linear ending values of the four clusters and are obtained by the K-means clustering technique and c_1, c_1, c_3 are thresholds for four congestion states corresponding to $T_1, T_2, \ldots T_6$. The expression of the proposed functional is represented as

$$S_1(v) = \begin{cases} 1, & v \le T_1 \\ \frac{T_2 - v}{T_2 - T_1}, & T_1 \le v \le T_2 \\ 0, & v \ge T_2 \end{cases} \quad S_2(v) = \begin{cases} \frac{v - T_1}{T_2 - T_1}, & T_1 \le v \le T_2 \\ 1, & T_2 \le v \le T_3 \\ \frac{T_4 - v}{T_4 - T_3}, & T_3 \le v \le T_4 \\ 0, & v \ge T_4 \end{cases}$$

$$S_3(v) = \begin{cases} \frac{v - T_3}{T_4 - T_3}, & T_3 \le v \le T_4 \\ 1, & T_4 \le v \le T_5 \\ \frac{T_6 - v}{T_6 - T_5}, & T_5 \le v \le T_6 \\ 0, & v \ge T_6 \end{cases} \quad S_4(v) = \begin{cases} \frac{v - T_5}{T_6 - T_5}, & T_5 \le v \le T_6 \\ 1, & v \ge T_6 \end{cases}$$

where $S_1(v)$, $S_2(v)$, $S_3(v)$, and $S_4(v)$, respectively, with the range of [0,1] are equivalant to traffic congestion states like serious congestion, congestion, medium, and low. Here, v_1 is the value of traffic speed, and its corresponding set $S = \{s_1, s_2, s_3, s_4\}$ correspond to serious congestion, congestion, medium, and low (called evaluation set). The value of proposed membership function r_{1j}, $j = 1, 2, \ldots, 4$ is determined for jth element of S with respect to traffic speed. Fuzzy evaluation set for a single parameter is expressed as in Eq. (12).

$$R = \begin{pmatrix} r_{11} & r_{12} & r_{13} & r_{14} \end{pmatrix} \tag{12}$$

After determining the fuzzy evaluation matrix, we can obtain the fuzzy comprehensive evaluation matrix B, which is represented in Eq. (13).

$$B = W \circ R = \begin{pmatrix} r_{11} & r_{12} & r_{13} & r_{14} \end{pmatrix} = (b_1, b_2, b_3, b_4) \tag{13}$$

where $\circ$ denote the fuzzy compositional operation and b_j denote the jth element of B. Applying max-membership principle, the biggest element $b = b_j$ of B represent traffic congestion state, as follow in Eq. (14)

$$b = b_j = \max(b_1, b_2, b_3, b_4). \tag{14}$$

4 Experiments

4.1 Performance Evaluation Metrics

This section describes the evaluation metrics, which have been utilized to determine the prediction performance and stability of applied models. During the experiments, metrics values were calculated from the predicted and the actual traffic speed. Equation (15)–(18) define the following metrics:

$$Mean\ squared\ error\ (MSE) = \frac{1}{n}\sum_{i=1}^{n}(\widehat{y}_i - y_i)^2 \quad (15)$$

$$Mean\ absolute\ error\ (MAE) = \frac{1}{n}\sum_{i=1}^{n}|\widehat{y}_i - y_i| \quad (16)$$

$$Mean\ absolute\ percentage\ error\ (MAPE) = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{\widehat{y}_i - y_i}{y_i}\right| 100\% \quad (17)$$

$$Coefficient\ of\ correlation\ (CC) = \frac{\sum_{i=1}^{n}(\widehat{y}_i - \bar{\widehat{y}})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(\widehat{y}_i - \bar{\widehat{y}})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}} \quad (18)$$

where y_i and $\widehat{y}_i$, respectively, are the actual and predicted speed during the ith time step from the test dataset; $\bar{y}$ is the mean of values of the y-variable and $\bar{\widehat{y}}$ is the mean of values of the $\widehat{y}$-variable; n is the total munber of predicted samples. Python 3.8.3 with the Keras 2.3.1, Tensorflow 2.0 was used for programming purpose.

4.2 Results and Discussion

To examine the prediction performance of applied NN models, each network was trained on 80% time intervals of datasets and tested (validated) on randomly selected 20% time intervals from the same datasets. All models were generated having similar architectures like input layers, output layers, and the number of the epoch which is shown in Table 2.

In a model, input layer receives the 9 features information (inputs) which includes time of the day, day of the week, and categorized vehicular speed and the output layer has one output i.e. traffic speed. In the process of information from inputs of the input layer into the output of the output layer, we need the activation function. The activation function produces a computer-understandable value. ReLU is a non-linear function and its range is between 0 and infinity.

Five different configurations of each network were designed for the prediction of traffic speed. Each configuration was having a single hidden layer and 150 epochs, while the number of hidden nodes were different. Each network performance was

Table 2 Structures of neural networks

Neural networks	Input feature	Output feature	Hidden layer	Epoch number	Activation function
MLP	Vehicular speed	Speed	1	150	Tansig
CFBP	Vehicular speed	Speed	1	150	Tansig
RNN	Vehicular speed	Speed	1	150	Tansig
LSTM	Vehicular speed	Speed	1	150	ReLU
CNN	Vehicular speed	Speed	1	150	ReLU
Conv-LSTM	Vehicular speed	Speed	1	150	ReLU

[b] Tansig—Hyperbolic tangent sigmoid, ReLU—Rectified linear unit

analyzed using metrics like MAE, MSE, MAPE, and CC. These models were trained by using the same training data of 5 min and 10 min intervals. Training and testing progress of Conv-LSTM model is shown in Fig. 5.

Prediction accuracy in 5 min and 10 min of interval for Conv-LSTM NN with other five neural network models MLP, CFBP, RNN, LSTM, and CNN is shown in the Tables 3 and 4. Best train of each model is highlighted in terms of hidden unit number, MAE, MSE, MAPE, and CC. Difference in the values of MAE, MSE, and MAPE between the best-performing model (LSTM) and Conv-LSTM are 0.2851, 0.2668, and 0.0124.

Table 3 shows that models MLP, CFBP, RNN, LSTM, CNN, and Conv-LSTM have highest coefficient of correlation values 97.47% (Train 4), 97.45% (Train 3), 96.46% (Train 1), 99.76% (Train 3), 99.49% (Train 2) and 99.98 % (Train 1) respectively for speed prediction in 5 min of interval. Table 4 shows that models MLP, CFBP, RNN, LSTM, CNN, and Conv-LSTM have highest coefficient of correlation values 97.12% (Train 1), 97.70% (Train 5), 97.59% (Train 1), 99.79% (Train 4), 99.88% (Train 4) and 99.93 % (Train 1), respectively, for speed prediction in 10 min of interval. Experimental results clearly show that the Conv-LSTM can capture both weekday and weekend time-series patterns. Furthermore, Conv-LSTM NN outperforms the other applied models in terms of accuracy and stability.

To analyze the congestion pattern for time intervals of testing dataset, developed congestion evaluation method was used. Congestion evaluation method has been described in Sect. 3.3. K-means clustering technique was used to determine the variables of congestion evaluation method which are shown in Table 5. In the values of the cluster, the first and second entries are the minimum and maximum values of the clusters, respectively.

Proposed congestion evaluation method classifies the traffic state (i.e., level of congestion) into four different categories namely low, medium, congestion, and serious congestion based upon traffic stream speed. Figure 6 represents a framework to

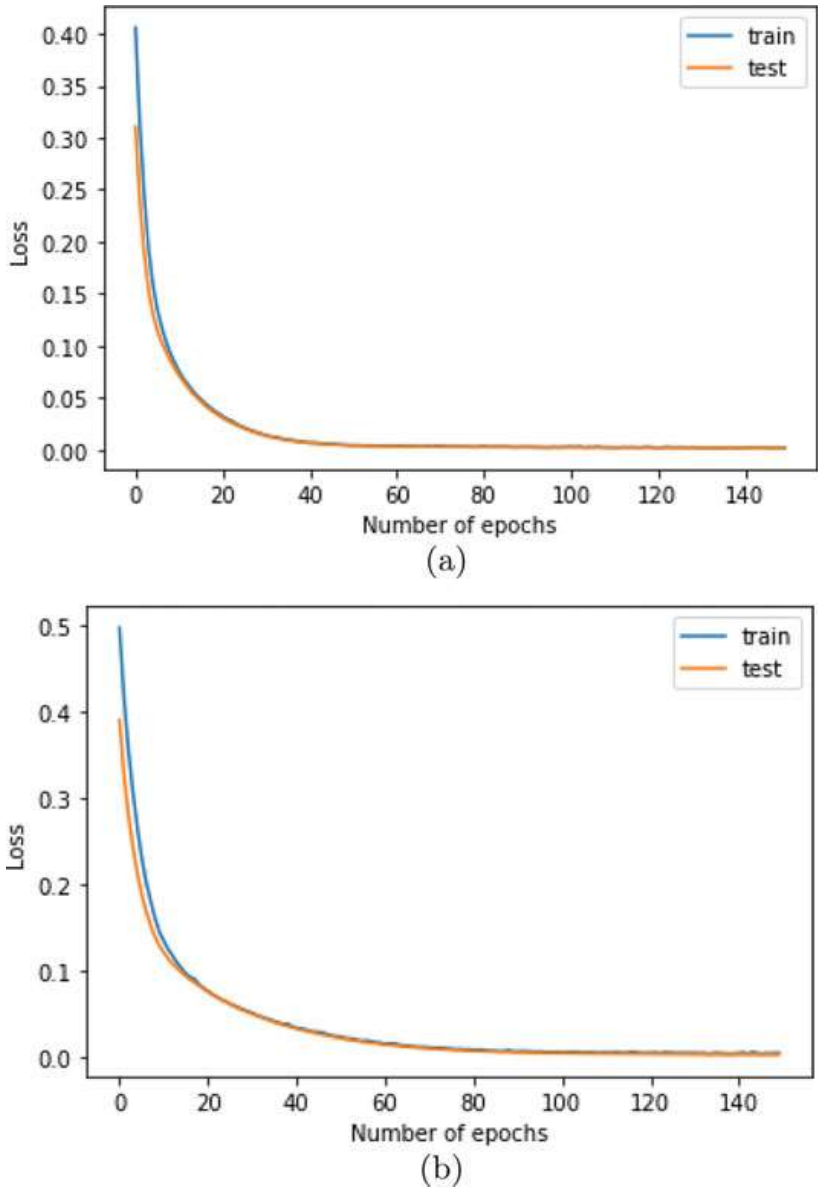


Fig. 5 Training progress and validation of Conv-LSTM model **a** for 5 min of intervals (Train 1) **b** for 10 min of intervals (Train 1)

predict the congestion state, in which the Conv-LSTM NN predicts traffic speed, and then developed evaluation method analyzes the type of traffic congestion state on the basis of the predicted speed. Table 6 shows some congested and non-congested time intervals using the developed congestion evaluation method. As per the time intervals, observed speed (average speed of categorized vehicular speeds) is taken from testing data, and predicted speed is the forecast speed corresponding to categorized vehicular speed (excluding average speed) from Conv-LSTM neural network. After that developed congestion evaluation method determines the congestion state according to traffic speed in terms of s_1, s_2, s_3, and s_4; where s_1, s_2, s_3, and s_4 represent serious

Table 3 Speed prediction for 5 min of intervals

	MLP					CFBP				
Model	Train 1	Train 2	Train 3	**Train 4**	Train 5	Train 1	Train 2	**Train 3**	Train 4	Train 5
HUN	10	25	40	**65**	80	10	25	**40**	65	80
MAE	1.1631	1.2130	1.1751	**1.1427**	1.2136	1.1300	1.2470	**1.1212**	1.2080	1.1781
MSE	4.7519	5.2385	4.8316	**4.7216**	5.0793	5.6828	5.9179	**5.5840**	5.6148	5.6079
MAPE	0.0649	0.0628	0.0637	**0.0623**	0.0755	0.0293	0.07668	**0.02346**	0.1319	0.0867
CC	0.9743	0.9675	0.9733	**0.9747**	0.9690	0.9744	0.9655	**0.9745**	0.9725	0.9736
	RNN					LSTM				
Model	**Train 1**	Train 2	Train 3	Train 4	Train 5	Train 1	Train 2	**Train 3**	Train 4	Train 5
HUN	**10**	25	40	65	80	10	25	**40**	65	80
MAE	**1.5213**	1.5607	1.5795	1.5926	1.5703	0.5507	0.5217	**0.3684**	0.4043	0.4447
MSE	**5.3077**	5.3228	5.3325	5.3380	5.3257	0.5429	2.0247	**0.2946**	0.7866	1.9515
MAPE	**0.0978**	0.1750	0.1391	0.1269	0.1179	0.0250	0.02221	**0.0165**	0.0181	0.0199
CC	**0.9646**	0.9634	0.9630	0.9624	0.9633	0.9958	0.9841	**0.9976**	0.9939	0.9852
	CNN					Conv-LSTM				
Model	Train 1	**Train 2**	Train 3	Train 4	Train 5	**Train 1**	Train 2	Train 3	Train 4	Train 5
HUN	10	**25**	40	65	80	**10**	25	40	65	80
MAE	0.9609	**0.5041**	0.6867	0.5051	0.9023	**0.0833**	0.2865	0.4987	0.0921	0.7470
MSE	1.7442	**0.6429**	2.5866	0.9664	1.7014	**0.0278**	0.1371	0.3616	0.0271	0.7685
MAPE	0.0439	**0.0223**	0.0309	0.0227	0.0557	**0.0041**	0.0136	0.0272	0.0054	0.0409
CC	0.9861	**0.9949**	0.9814	0.9934	0.9927	**0.9998**	0.9998	0.9998	0.9998	0.9997

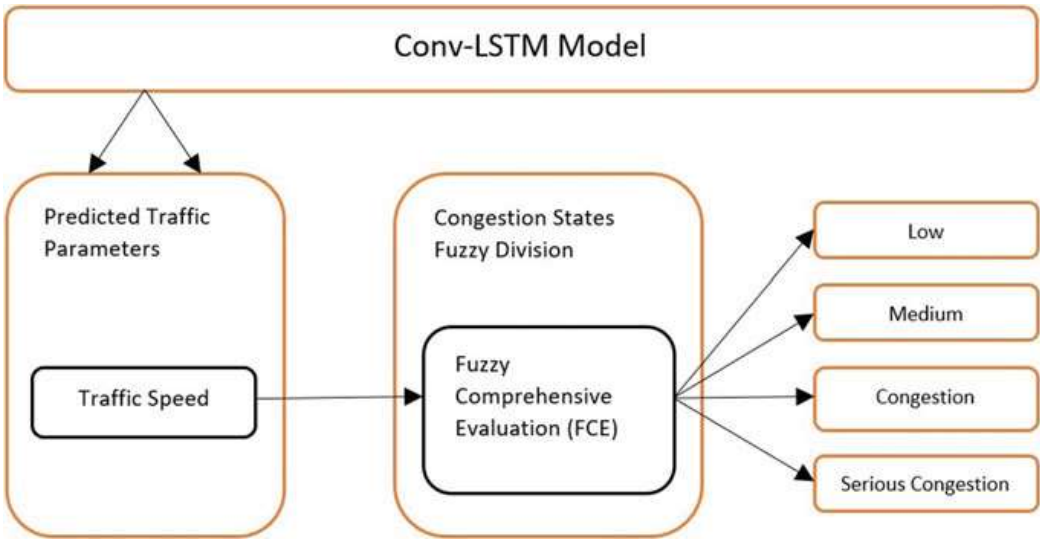


Fig. 6 Flow chart of traffic congestion state prediction with Conv-LSTM

Table 4 Speed prediction for 10 min of intervals

	MLP					CFBP				
Model	**Train 1**	Train 2	Train 3	Train 4	Train 5	Train 1	Train 2	Train 3	Train 4	**Train 5**
HUN	**10**	25	40	65	80	10	25	40	65	**80**
MAE	**1.0239**	1.0438	1.0430	1.0675	1.0535	0.9908	1.0345	1.0551	1.0256	**0.9708**
MSE	**3.8924**	4.1762	4.1719	4.1937	4.1788	4.2053	4.2750	4.0466	3.9742	**3.8195**
MAPE	**0.0718**	0.0762	0.1073	0.1266	0.0812	0.1217	0.0869	0.0621	0.04963	**0.02848**
CC	**0.9712**	0.9661	0.9663	0.9659	0.9662	0.9767	0.9720	0.9658	0.9711	**0.9770**
	RNN					LSTM				
Model	**Train 1**	Train 2	Train 3	Train 4	Train 5	Train 1	Train 2	Train 3	**Train 4**	Train 5
HUN	**10**	25	40	65	80	10	25	40	**65**	80
MAE	**1.0525**	1.1281	1.2525	1.3118	1.3263	0.9589	0.4232	0.3988	**0.3373**	0.3612
MSE	**3.6755**	3.7780	3.9629	4.0637	4.1008	1.7984	0.4566	0.5137	**0.2785**	0.2642
MAPE	**0.0994**	0.1027	0.0996	0.1185	1.1441	0.0478	0.0196	0.0179	**0.0159**	0.0187
CC	**0.9759**	0.9742	0.9710	0.9690	0.9687	0.9836	0.9958	0.9954	**0.9979**	0.9978
	CNN					Conv-LSTM				
Model	Train 1	Train 1	Train 3	**Train 4**	Train 5	**Train 1**	Train 2	Train 3	Train 4	Train 5
HUN	10	25	40	**65**	80	**10**	25	40	65	80
MAE	0.6871	0.3371	0.3468	**0.3013**	0.3261	**0.1167**	0.2827	0.2941	0.3428	0.1525
MSE	0.8405	0.2885	0.2465	**0.2210**	0.2382	**0.1002**	0.1945	0.1946	0.1956	0.1005
MAPE	0.0379	0.0151	0.0153	**0.0144**	0.0186	**0.0069**	0.0162	0.015	0.0205	0.0074
CC	0.9924	0.9982	0.9985	**0.9988**	0.9988	**0.9993**	0.9992	0.9992	0.9992	0.9993

[c] HUN—Hidden unit number, MAE—Mean absolute error, MSE—mean square error, MAPE—Mean absolute percentage error, and CC—Coefficient of correlation

Table 5 Linear values of four clusters

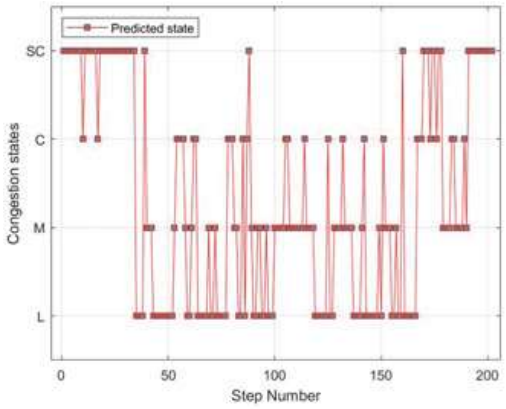
No of cluster	Values of cluster	Variable
1	07.97, 15.41	T_0, T_1
2	15.45, 20.88	T_2, T_3
3	20.98, 28.21	T_4, T_5
4	28.39, 43.46	T_6, T_7

congestion (SC), congestion (C), medium (M), and low (L), respectively. Testing datasets of 5 min intervals and 10 min intervals were used to generate Fig. 7 and it shows the comparison between predicted congestion state by predicted speed and estimated congestion state by observed speed. Figure 7 confirms that the predicted congestion state and observed congested state is almost the same. It affirms that the proposed method can be applied successfully to the prediction of traffic congestion state in non-homogeneous traffic.

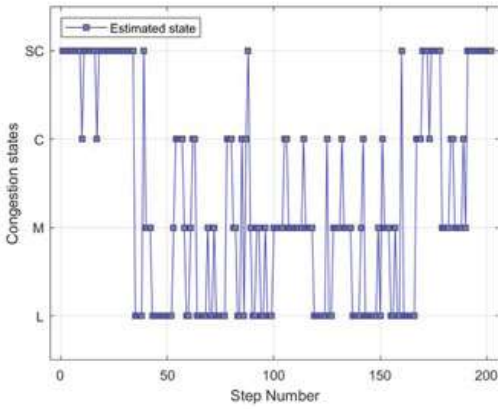
Table 6 Congestion class prediction

Time intervals	Datetime	Obs. speed	b value	Obs. state	Pred. speed	b value	Pred. state
1	16-02-2021 10:25	13.92	1	s_1	14.12	1	s_1
2	16-02-2021 10:55	27.05	1	s_3	26.96	1	s_3
3	16-02-2021 16:25	23.92	1	s_3	23.90	1	s_3
4	16-02-2021 16:55	22.31	1	s_3	22.30	1	s_3
5	16-02-2021 17:25	30.29	1	s_4	29.92	1	s_4

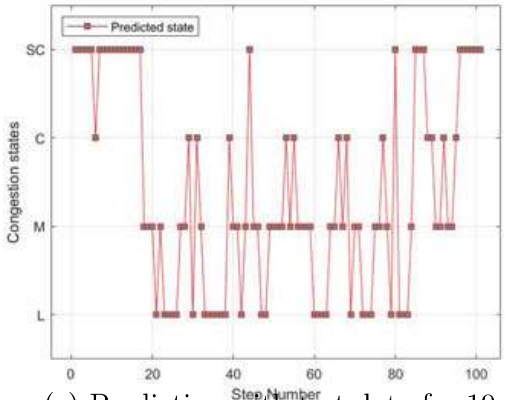
[d] Obs.—Observed, Pred.—Predicted, s_1—Serious congestion, s_2—Congestion, s_3—Medium, s_4—Low



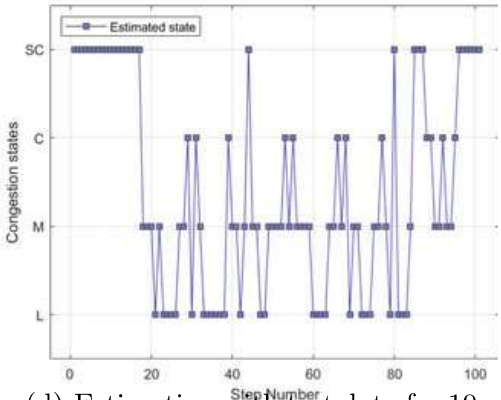
(a) Prediction with test data for 5 minutes



(b) Estimation with test data for 5 minutes



(c) Prediction with test data for 10 minutes



(d) Estimation with test data for 10 minutes

Fig. 7 Comparision between predicted and estimated congestion states

5 Conclusion

In this study, six neural network architectures were constructed with required properties having the same training and testing datasets. Each lag of datasets contains nine input features (e.g., Time of the day, day of the week, categorized vehicular average speed) and one output (e.g., average stream speed or traffic speed). For short-term traffic speed prediction, the performance of the networks was evaluated. The prediction accuracy of these networks was measured by metrics such as MSE, MAE, and MAPE, and stability by the coefficient of correlation (CC). Study results suggest that the Conv-LSTM neural network can be successfully applied for short-term prediction of traffic speed. In this article, our second aim was traffic congestion state prediction. For this purpose, we developed a congestion evaluation method based on the traffic speed using fuzzy comprehensive evaluation scheme. Finally, the developed congestion evaluation method was combined with the Conv-LSTM neural network which then predicted the traffic congestion state using categorized vehicular speed data. Study results clearly demonstrate that the Conv-LSTM neural network and congestion evaluation method can be successfully applied for traffic congestion prediction.

More datasets are required to test the efficiency and robustness of the proposed traffic congestion prediction method under diverse traffic conditions. Due to COVID-19-induced conditions, it was difficult for us to collect more amount of data. Future studies will focus on the collection of more data so that the proposed traffic congestion prediction method can be tested under different types of traffic conditions.

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Conflict of Interest Authors declare that they have no conflict of interest.

References

1. Bai M, Lin Y, Ma M, Wang P, Duan L (2021) Prepct: traffic congestion prediction in smart cities with relative position congestion tensor. Neurocomputing 444:147–157
2. Bengio Y, Simard P, Frasconi P (1994) Learning long-term dependencies with gradient descent is difficult. IEEE Trans Neural Netw 5:157–166
3. Bhatia J, Dave R, Bhayani H, Tanwar S, Nayyar A (2020) SDN-based real-time urban traffic analysis in vanet environment. Comput Commun 149:162–175
4. Boukerche A, Wang J (2020) A performance modeling and analysis of a novel vehicular traffic flow prediction system using a hybrid machine learning-based model. Ad Hoc Netw 106:102224
5. Chakraborty P, Adu-Gyamfi YO, Poddar S, Ahsani V, Sharma A, Sarkar S (2018) Traffic congestion detection from camera images using deep convolution neural networks. Transp Res Rec 2672:222–231

6. CSIR-CRRI: the council of scientific and industrial research-central road research institute conducted the study (2016). http://www.hindustantimes.com/delhi/poor-upkeep-encroachment-cause-jams-on-internal-delhi-roads-says-study/story-UlwR5FYUrmW2EUjtqJQtcK.html
7. CSIR-CRRI: the council of scientific and industrial research-central road research institute conducted the study (2017). http://www.hindustantimes.com/delhi-news/people-pay-more-for-extra-fuel-consumption-while-driving-on-delhi-s-roads/story-68rK37RfWJHX30UpFBFqWM.html
8. Fu R, Zhang Z, Li L (2016) Using LSTM and GRU neural network methods for traffic flow prediction. In: 2016 31st youth academic annual conference of chinese association of automation (YAC). IEEE, pp 324–328
9. Gu Y, Lu W, Qin L, Li M, Shao Z (2019) Short-term prediction of lane-level traffic speeds: a fusion deep learning model. Transp Res Part C: Emerg Technol 106:1–16
10. Hochreiter S, Schmidhuber J (1997) Long short-term memory. Neural Comput 9:1735–1780
11. Kong X, Xu Z, Shen G, Wang J, Yang Q, Zhang B (2016) Urban traffic congestion estimation and prediction based on floating car trajectory data. Future Gener Comput Syst 61:97–107
12. Kumar M, Kumar K, Das P (2021) Study on road traffic congestion: a review. In: Recent Trends in Communication and Electronics, pp 230–240
13. Kumar SV (2017) Traffic flow prediction using Kalman filtering technique. Procedia Eng 187:582–587
14. Kumar SV, Vanajakshi L (2015) Short-term traffic flow prediction using seasonal ARIMA model with limited input data. Eur Transp Res Rev 7:1–9
15. LeCun Y, Boser B, Denker JS, Henderson D, Howard RE, Hubbard W, Jackel LD (1989) Backpropagation applied to handwritten zip code recognition. Neural Comput 1:541–551
16. Li L, Qin L, Qu X, Zhang J, Wang Y, Ran B (2019) Day-ahead traffic flow forecasting based on a deep belief network optimized by the multi-objective particle swarm algorithm. Knowl-Based Syst 172:1–14
17. Liu Y, Zheng H, Feng X, Chen Z (2017) Short-term traffic flow prediction with conv-lstm. In: 2017 9th international conference on wireless communications and signal processing (WCSP). IEEE, pp. 1–6. https://doi.org/10.1109/WCSP.2017.8171119
18. Lu W, Rui Y, Ran B (2020) Lane-level traffic speed forecasting: a novel mixed deep learning model. IEEE Trans Intell Transp Syst 1–12. https://doi.org/10.1109/TITS.2020.3038457
19. Ma X, Tao Z, Wang Y, Yu H, Wang Y (2015) Long short-term memory neural network for traffic speed prediction using remote microwave sensor data. Transp Res Part C: Emerg Technol 54:187–197
20. Mahmuda Akhtar SM (2021) A review of traffic congestion prediction using artificial intelligence. J Adv Transp 2021. https://doi.org/10.1155/2021/8878011
21. Mohanty S, Pozdnukhov A, Cassidy M (2020) Region-wide congestion prediction and control using deep learning. Transp Res Part C: Emerg Technol 116:102624
22. Ranjan N, Bhandari S, Zhao HP, Kim H, Khan P (2020) City-wide traffic congestion prediction based on CNN, LSTM and transpose CNN. IEEE Access 8:81606–81620
23. Rao M, Rao KR (2016) Identification of traffic congestion on urban arterials for heterogeneous traffic. Transp Probl 11. https://doi.org/10.20858/tp.2016.11.3.13
24. Su H, Zhang L, Yu S (2007) Short-term traffic flow prediction based on incremental support vector regression. In: Third international conference on natural computation (ICNC 2007), vol 1. IEEE, pp 640–645
25. Tian Y, Zhang K, Li J, Lin X, Yang B (2018) LSTM-based traffic flow prediction with missing data. Neurocomputing 318:297–305
26. Tu Y, Lin S, Qiao J, Liu B (2021) Deep traffic congestion prediction model based on road segment grouping. Appl Intell 1–23
27. Williams BM (2001) Multivariate vehicular traffic flow prediction: evaluation of ARIMAX modeling. Transp Res Rec 1776:194–200
28. Wu Y, Tan H, Qin L, Ran B, Jiang Z (2018) A hybrid deep learning based traffic flow prediction method and its understanding. Transp Res Part C: Emerg Technol 90:166–180

29. Xiao Z, Xia S, Gong K, Li D (2012) The trapezoidal fuzzy soft set and its application in MCDM. Appl Math Model 36(12):5844–5855
30. Zaki JF, Ali-Eldin A, Hussein SE, Saraya SF, Areed FF (2020) Traffic congestion prediction based on hidden Markov models and contrast measure. Ain Shams Eng J 11:535–551